

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : Case Study (Medium)

Assessment Tool Weightage: 30%

QUESTIONS Marks: 50			Total
Q.No	Question	Bloom's Level	Marks
1	Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.	BL2 – Understand	5
2	A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.	BL3 – Apply	5
3	Case study: A university stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.	BL3 – Apply	5

4	Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.	BL3 – Apply	5
5	Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF.	BL2 – Understand	5
6	Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.	BL3 – Apply	5
7	Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.	BL2 – Understand	5
8	A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.	BL3 – Apply	5
9	Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification.	BL3 – Apply	5
10	Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.	BL3 – Apply	5

Code of Conduct:

I D. Gnanadeepthi(192411123) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D. Gnanadeepthi.

ANSWERS

1.

Given Relation

Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price)

Functional Dependencies

The functional dependencies are:

- $\text{OrderID} \rightarrow \text{CustID}$
- $\text{CustID} \rightarrow \text{CustName}$
- $\text{ProdID} \rightarrow \text{ProdName, Price}$
- $(\text{OrderID, ProdID}) \rightarrow \text{Qty}$

Since an order can contain multiple products, the candidate key is:

(OrderID, ProdID)

First Normal Form (1NF)

The relation is in **1NF** because all attributes contain atomic values and there are no repeating groups.

However, the relation is not in 2NF because some attributes depend only on part of the composite key.

Second Normal Form (2NF)

Partial dependencies are:

- $\text{OrderID} \rightarrow \text{CustID}$
- $\text{ProdID} \rightarrow \text{ProdName, Price}$

To remove these partial dependencies, decompose the relation into:

Orders(OrderID, CustID)

Products(ProdID, ProdName, Price)

OrderDetails(OrderID, ProdID, Qty)

4. Third Normal Form (3NF)

There is a transitive dependency:

$\text{OrderID} \rightarrow \text{CustID} \rightarrow \text{CustName}$

Therefore, CustName should be separated into a Customer relation.

The final 3NF relations are:

Customers(CustID, CustName)

- Primary Key: CustID

Orders(OrderID, CustID)

- Primary Key: OrderID
- Foreign Key: CustID

Products(ProdID, ProdName, Price)

- Primary Key: ProdID

OrderDetails(OrderID, ProdID, Qty)

- Composite Primary Key: (OrderID, ProdID)
- Foreign Keys: OrderID, ProdID

Final 3NF Schema

Relation	Attributes	Primary Key
Customers	CustID, CustName	CustID
Orders	OrderID, CustID	OrderID
Products	ProdID, ProdName, Price	ProdID
OrderDetails	OrderID, ProdID, Qty	(OrderID, ProdID)

2.

Given Relation

Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName)

Functional Dependencies

Assume:

- $PID \rightarrow PName, DocID, WardNo$

- $\text{DocID} \rightarrow \text{DocName}, \text{Specialization}$
- $\text{WardNo} \rightarrow \text{WardName}$

The candidate key is:

PID

because PID uniquely identifies each patient.

Anomalies

The original relation has the following anomalies:

Update Anomaly:

If a doctor's specialization changes, the information must be updated for every patient treated by that doctor. Missing an update can create inconsistent data.

Insertion Anomaly:

A new doctor cannot be added to the database unless the doctor is assigned to a patient.

Deletion Anomaly:

If the last patient assigned to a particular doctor is deleted, information about that doctor may also be lost.

Similar redundancy occurs with ward information because WardNo determines WardName .

BCNF Analysis

A relation is in **BCNF** if, for every non-trivial functional dependency $\mathbf{X} \rightarrow \mathbf{Y}$, $\mathbf{X}$ is a superkey.

In the original relation:

- $\text{PID} \rightarrow \text{PName}, \text{DocID}, \text{WardNo} \rightarrow \text{PID}$ is a key, so this is valid.
- $\text{DocID} \rightarrow \text{DocName}, \text{Specialization} \rightarrow \text{DocID}$ is not a superkey, so it violates BCNF.
- $\text{WardNo} \rightarrow \text{WardName} \rightarrow \text{WardNo}$ is not a superkey, so it also violates BCNF.

Therefore, the original relation is not in BCNF.

Decomposition into BCNF

Decompose the relation based on the violating dependencies.

Relation 1: Patient

Patient(PID, PName, DocID, WardNo)

- Primary Key: PID
- Foreign Keys: DocID, WardNo

Relation 2: Doctor

Doctor(DocID, DocName, Specialization)

- Primary Key: DocID

Relation 3: Ward

Ward(WardNo, WardName)

- Primary Key: WardNo

Final BCNF Schema

Relation	Attributes	Primary Key
Patient	PID, PName, DocID, WardNo	PID
Doctor	DocID, DocName, Specialization	DocID
Ward	WardNo, WardName	WardNo

3.

Given Relation

Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade)

Functional Dependencies

The functional dependencies are:

- $SID \rightarrow SName$
- $CourseID \rightarrow CourseName, Faculty$
- $(SID, CourseID) \rightarrow Grade$

Since a student can enroll in multiple courses and a course can have multiple students, the candidate key is:

$(SID, CourseID)$

1NF Analysis

The relation is in First Normal Form (1NF) because all attributes contain atomic values and there are no repeating groups.

However, it is not in 2NF because there are partial dependencies.

Partial Dependencies

The composite key is (SID, CourseID).

The following attributes depend only on part of the key:

- $SID \rightarrow SName$
- $CourseID \rightarrow CourseName, Faculty$

These are partial dependencies and violate 2NF.

2NF Decomposition

To remove the partial dependencies, decompose the relation into three relations:

Student Relation

Student(SID, SName)

- Primary Key: SID

Course Relation

Course(CourseID, CourseName, Faculty)

- Primary Key: CourseID

Enrollment Relation

Enrollment(SID, CourseID, Grade)

- Composite Primary Key: (SID, CourseID)
- SID is a Foreign Key referencing Student.
- CourseID is a Foreign Key referencing Course.

Final 2NF Schema

Relation	Attributes	Primary Key
Student	SID, SName	SID
Course	CourseID, CourseName, Faculty	CourseID
Enrollment	SID, CourseID, Grade	(SID, CourseID)

4.

Case Study

Given the airline booking relation:

Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date)

Normalize the relation up to Boyce-Codd Normal Form (BCNF).

1. Functional Dependencies

Assume the following functional dependencies:

- $\text{PassID} \rightarrow \text{PassName}$
- $\text{FlightNo} \rightarrow \text{DeptCity, ArrCity}$
- $(\text{FlightNo, Date, Seat}) \rightarrow \text{PassID}$
- $(\text{FlightNo, PassID, Date}) \rightarrow \text{Seat}$

The candidate keys are:

- $(\text{FlightNo, Date, Seat})$
- $(\text{FlightNo, PassID, Date})$

2. First Normal Form (1NF)

The relation is in **1NF** because all attributes contain atomic values and there are no repeating groups.

3. Second Normal Form (2NF)

There are partial dependencies:

- $\text{PassID} \rightarrow \text{PassName}$
- $\text{FlightNo} \rightarrow \text{DeptCity, ArrCity}$

These attributes depend only on part of the composite keys.

Decompose the relation into:

Passenger(PassID, PassName)

Flight(FlightNo, DeptCity, ArrCity)

Booking(FlightNo, PassID, Seat, Date)

4. Third Normal Form (3NF)

In the decomposed relations:

- In Passenger, PassID is the key.
- In Flight, FlightNo is the key.
- In Booking, the candidate keys are $(\text{FlightNo, Date, Seat})$ and $(\text{FlightNo, PassID, Date})$.

There are no transitive dependencies among non-key attributes.

Therefore, the relations satisfy 3NF.

5. BCNF Analysis

A relation is in BCNF if, for every non-trivial functional dependency $X \rightarrow Y$, X must be a superkey.

Passenger

$\text{PassID} \rightarrow \text{PassName}$

PassID is the primary key, so Passenger is in BCNF.

Flight

$\text{FlightNo} \rightarrow \text{DeptCity}, \text{ArrCity}$

FlightNo is the primary key, so Flight is in BCNF.

Booking

The determinants are:

- (FlightNo, Date, Seat)
- (FlightNo, PassID, Date)

Both are candidate keys and therefore superkeys.

Hence, Booking is also in BCNF.

6. Final BCNF Schema

Relation	Attributes	Primary Key
Passenger	PassID, PassName	PassID
Flight	FlightNo, DeptCity, ArrCity	FlightNo
Booking	FlightNo, PassID, Seat, Date	(FlightNo, Date, Seat)

Alternate candidate key of Booking: (FlightNo, PassID, Date)

5.

Case Study: HR System of a Multinational Company

Consider the following HR relation:

Employee(EmpID, EmpName, DeptID, DeptName, ManagerID, ManagerName, CountryID, CountryName, JobID, JobTitle, Salary)

The objective is to identify the functional and transitive dependencies and normalize the relation up to Third Normal Form (3NF).

1. Functional Dependencies

The main functional dependencies are:

- $\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}, \text{ManagerID}, \text{CountryID}, \text{JobID}, \text{Salary}$
- $\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$
- $\text{ManagerID} \rightarrow \text{ManagerName}$
- $\text{CountryID} \rightarrow \text{CountryName}$
- $\text{JobID} \rightarrow \text{JobTitle}$

The candidate key is:

EmpID

because each employee is uniquely identified by EmpID.

2. Transitive Dependencies

A transitive dependency occurs when a non-key attribute determines another non-key attribute.

The following transitive dependencies exist:

- $\text{EmpID} \rightarrow \text{DeptID} \rightarrow \text{DeptName}$
- $\text{EmpID} \rightarrow \text{DeptID} \rightarrow \text{ManagerID}$
- $\text{EmpID} \rightarrow \text{ManagerID} \rightarrow \text{ManagerName}$
- $\text{EmpID} \rightarrow \text{CountryID} \rightarrow \text{CountryName}$
- $\text{EmpID} \rightarrow \text{JobID} \rightarrow \text{JobTitle}$

For example:

$\text{EmpID} \rightarrow \text{DeptID}$ and $\text{DeptID} \rightarrow \text{DeptName}$

Therefore:

$\text{EmpID} \rightarrow \text{DeptName}$

This is a transitive dependency.

3. First Normal Form (1NF)

The relation is in **1NF** because all attributes contain atomic values and there are no repeating groups.

4. Second Normal Form (2NF)

Since **EmpID** is a single-attribute candidate key, there are no partial dependencies.

Therefore, the relation is already in **2NF**.

However, it contains several transitive dependencies, so it is not in 3NF.

5. Third Normal Form (3NF)

To eliminate transitive dependencies, decompose the relation into the following tables.

Employee

Employee(EmpID, EmpName, DeptID, ManagerID, CountryID, JobID, Salary)

- Primary Key: EmpID
- Foreign Keys: DeptID, ManagerID, CountryID, JobID

Department

Department(DeptID, DeptName, ManagerID)

- Primary Key: DeptID
- ManagerID is a Foreign Key.

Manager

Manager(ManagerID, ManagerName)

- Primary Key: ManagerID

Country

Country(CountryID, CountryName)

- Primary Key: CountryID

Job

Job(JobID, JobTitle)

- Primary Key: JobID

6. Final 3NF Schema

Relation	Attributes	Primary Key
Employee	EmpID, EmpName, DeptID, ManagerID, CountryID, JobID, Salary	EmpID
Department	DeptID, DeptName, ManagerID	DeptID
Manager	ManagerID, ManagerName	ManagerID
Country	CountryID, CountryName	CountryID
Job	JobID, JobTitle	JobID

7. Advantages of 3NF

The decomposition:

- Removes transitive dependencies.
- Reduces data redundancy.
- Prevents update anomalies.
- Prevents insertion and deletion anomalies.
- Makes employee, department, manager, country, and job information easier to maintain.
- Maintains data integrity using primary and foreign keys.

6.

Case Study: Library Management System

Given relation:

Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate)

The objective is to identify Multivalued Dependencies (MVDs) and decompose the relation into Fourth Normal Form (4NF).

1. Functional Dependencies

The important functional dependencies are:

- $\text{MemberID} \rightarrow \text{MemberName}$
- $\text{BookID} \rightarrow \text{Title, Author, Genre}$

The candidate key for the borrowing information is:

(MemberID, BookID, BorrowDate)

2. Multivalued Dependencies

Suppose a member can borrow multiple books and each book can have multiple authors independently of the member's borrowing details.

The important multivalued dependencies are:

- $\text{MemberID} \twoheadrightarrow \text{BookID}$
- $\text{MemberID} \twoheadrightarrow \text{BorrowDate}$

This means that the books borrowed by a member and the borrowing dates can be independent multivalued facts.

These independent multivalued dependencies can cause unnecessary combinations and redundancy.

3. Why the Relation Violates 4NF

A relation is in 4NF if, for every non-trivial MVD $X \twoheadrightarrow Y$, X is a superkey.

Here:

$\text{MemberID} \twoheadrightarrow \text{BookID}$

But MemberID is not a superkey of the original Library relation.

Therefore, the relation violates 4NF.

4. Decomposition

First separate member information:

Member(MemberID, MemberName)

Then separate book information:

Book(BookID, Title, Author, Genre)

Finally, keep the borrowing relationship:

Borrow(MemberID, BookID, BorrowDate)

5. Final 4NF Relations

Relation	Attributes	Primary Key
Member	MemberID, MemberName	MemberID
Book	BookID, Title, Author, Genre	BookID
Borrow	MemberID, BookID, BorrowDate	(MemberID, BookID, BorrowDate)

6. 4NF Verification

- **Member:** $\text{MemberID} \rightarrow \text{MemberName}$, so it satisfies 4NF.
- **Book:** $\text{BookID} \rightarrow \text{Title}, \text{Author}, \text{Genre}$, so it satisfies 4NF.
- **Borrow:** The independent multivalued facts are separated, so no non-trivial MVD with a non-superkey determinant remains.

Hence, all the resulting relations satisfy **4NF**.

7.

Case Study: Telecom Billing System

Consider the following telecom billing relation:

Billing(BillID, CustomerID, CustomerName, PhoneNo, PlanID, PlanName, MonthlyCharge, CallID, CallDate, CallDuration, CallCharge)

The objective is to identify all candidate keys, primary keys, and functional dependencies before normalization.

1. Functional Dependencies

The functional dependencies are:

1. $\text{BillID} \rightarrow \text{CustomerID}, \text{PlanID}$
2. $\text{CustomerID} \rightarrow \text{CustomerName}, \text{PhoneNo}$
3. $\text{PlanID} \rightarrow \text{PlanName}, \text{MonthlyCharge}$
4. $\text{CallID} \rightarrow \text{BillID}, \text{CallDate}, \text{CallDuration}, \text{CallCharge}$

Thus, the following dependencies also exist transitively:

- $\text{BillID} \rightarrow \text{CustomerID} \rightarrow \text{CustomerName}, \text{PhoneNo}$
- $\text{BillID} \rightarrow \text{PlanID} \rightarrow \text{PlanName}, \text{MonthlyCharge}$

2. Candidate Key

Assume that each call record has a unique CallID.

Therefore:

CallID → BillID, CustomerID, CustomerName, PhoneNo, PlanID, PlanName, MonthlyCharge, CallDate, CallDuration, CallCharge

Hence, CallID uniquely identifies every record.

Candidate Key: CallID

3. Primary Key

Since there is one candidate key, it can be selected as the primary key:

Primary Key = CallID

4. Summary of Functional Dependencies

Determinant Dependent Attributes

BillID CustomerID, PlanID

CustomerID CustomerName, PhoneNo

PlanID PlanName, MonthlyCharge

CallID BillID, CallDate, CallDuration, CallCharge

5. Candidate Key and Primary Key

Key Type	Attribute
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Candidate Key	CallID
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Primary Key	CallID
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8.

Case Study: Retail Inventory System

Given Relation:

Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse)

The objective is to apply complete normalization up to Third Normal Form (3NF).

1. Functional Dependencies

The functional dependencies are:

- **PID → PName, SupplierID, Category, Price, Warehouse**
- **SupplierID → SupplierName**

Therefore, there is a transitive dependency:

PID → SupplierID → SupplierName

The candidate key is:

PID

2. First Normal Form (1NF)

The relation is in **1NF** because:

- All attributes contain atomic values.
- There are no repeating groups.
- Each product is uniquely identified by PID.

3. Second Normal Form (2NF)

Since the candidate key **PID** consists of a single attribute, there can be no partial dependency.

Therefore, the relation is already in **2NF**.

4. Third Normal Form (3NF)

The relation is not in 3NF because of the transitive dependency:

PID → SupplierID → SupplierName

SupplierName depends on SupplierID rather than directly on the primary key PID.

To remove this dependency, decompose the relation.

Product

Product(PID, PName, SupplierID, Category, Price, Warehouse)

- Primary Key: PID
- SupplierID is a Foreign Key.

Supplier

Supplier(SupplierID, SupplierName)

- Primary Key: SupplierID

5. Final Normalized Schema

Relation Attributes

Primary Key

Product PID, PName, SupplierID, Category, Price, Warehouse PID

Supplier SupplierID, SupplierName

SupplierID

6. Normalization Summary

UNF $\rightarrow$ 1NF $\rightarrow$ 2NF $\rightarrow$ 3NF

- **1NF:** Atomic attributes and no repeating groups.
- **2NF:** Already satisfied because PID is a single-attribute key.
- **3NF:** Transitive dependency is removed by separating Supplier information.

9.

Since no specific attributes are provided in the question, consider the following typical poorly normalized School ERP relation:

SchoolERP(StudentID, StudentName, ClassID, ClassName, TeacherID, TeacherName, SubjectID, SubjectName, ExamID, ExamName, Marks)

1. Functional Dependencies

The important functional dependencies are:

- StudentID $\rightarrow$ StudentName, ClassID
- ClassID $\rightarrow$ ClassName, TeacherID
- TeacherID $\rightarrow$ TeacherName
- SubjectID $\rightarrow$ SubjectName
- ExamID $\rightarrow$ ExamName
- (StudentID, SubjectID, ExamID) $\rightarrow$ Marks

Therefore, the candidate key for the original relation is:

(StudentID, SubjectID, ExamID)

2. Problems in the Original Schema

The poorly normalized relation causes:

Update Anomaly:

If a teacher's name changes, it may have to be updated in many student records.

Insertion Anomaly:

A new subject or teacher cannot be stored unless there is a corresponding student record.

Deletion Anomaly:

Deleting the last student enrolled in a class may accidentally remove information about that class or its teacher.

Data Redundancy:

Student, class, teacher, subject, and examination information is repeatedly stored.

3. First Normal Form (1NF)

The relation satisfies 1NF if:

- Every attribute contains atomic values.
- There are no repeating groups.
- Each row represents one student's marks for one subject and examination.

However, it contains partial and transitive dependencies.

4. Second Normal Form (2NF)

The composite key is:

(StudentID, SubjectID, ExamID)

Partial dependencies include:

- StudentID $\rightarrow$ StudentName, ClassID
- SubjectID $\rightarrow$ SubjectName
- ExamID $\rightarrow$ ExamName

To remove these dependencies, decompose the relation into:

Student(StudentID, StudentName, ClassID)

Subject(SubjectID, SubjectName)

Exam(ExamID, ExamName)

Result(StudentID, SubjectID, ExamID, Marks)

5. Third Normal Form (3NF)

There are transitive dependencies:

StudentID $\rightarrow$ ClassID $\rightarrow$ ClassName, TeacherID

and

StudentID $\rightarrow$ ClassID $\rightarrow$ TeacherID $\rightarrow$ TeacherName

Therefore, class and teacher information must be separated.

The 3NF relations become:

Student(StudentID, StudentName, ClassID)

Class(ClassID, ClassName, TeacherID)

Teacher(TeacherID, TeacherName)

Subject(SubjectID, SubjectName)

Exam(ExamID, ExamName)

Result(StudentID, SubjectID, ExamID, Marks)

6. BCNF Analysis

A relation is in **BCNF** if, for every non-trivial functional dependency $X \rightarrow Y$, X is a superkey.

Student

StudentID $\rightarrow$ StudentName, ClassID

StudentID is the primary key, so Student is in BCNF.

Class

ClassID $\rightarrow$ ClassName, TeacherID

ClassID is the primary key, so Class is in BCNF.

Teacher

TeacherID $\rightarrow$ TeacherName

TeacherID is the primary key, so Teacher is in BCNF.

Subject

SubjectID $\rightarrow$ SubjectName

SubjectID is the primary key, so Subject is in BCNF.

Exam

ExamID $\rightarrow$ ExamName

ExamID is the primary key, so Exam is in BCNF.

Result

(StudentID, SubjectID, ExamID) $\rightarrow$ Marks

The determinant is the composite primary key, so Result is in BCNF.

7. Final BCNF Schema

Relation	Attributes	Primary Key
Student	StudentID, StudentName, ClassID	StudentID
Class	ClassID, ClassName, TeacherID	ClassID
Teacher	TeacherID, TeacherName	TeacherID
Subject	SubjectID, SubjectName	SubjectID
Exam	ExamID, ExamName	ExamID
Result	StudentID, SubjectID, ExamID, Marks (StudentID, SubjectID, ExamID)	

8. Justification

The final design satisfies **BCNF** because every determinant in each relation is a candidate key or superkey.

The normalization:

1NF → 2NF → 3NF → BCNF

removes:

- Repeating and non-atomic data.
- Partial dependencies.
- Transitive dependencies.
- Data redundancy.
- Update, insertion, and deletion anomalies.

The final BCNF design also maintains relationships through foreign keys:

- Student.ClassID → Class.ClassID
- Class.TeacherID → Teacher.TeacherID
- Result.StudentID → Student.StudentID
- Result.SubjectID → Subject.SubjectID
- Result.ExamID → Exam.ExamID

10.

Consider the following relation:

Streaming(MovieID, MovieTitle, ActorID, ActorName, DirectorID, DirectorName, PlatformID, PlatformName)

The objective is to identify **join dependencies** and decompose the relation into **Fifth Normal Form (5NF)** with lossless-join verification.

1. Functional Dependencies

The basic functional dependencies are:

- **MovieID $\rightarrow$ MovieTitle**
- **ActorID $\rightarrow$ ActorName**
- **DirectorID $\rightarrow$ DirectorName**
- **PlatformID $\rightarrow$ PlatformName**

The remaining attributes describe relationships between a movie, its actors, directors, and streaming platforms.

2. Join Dependency

Assume that the participation of an actor, director, and platform for a movie are independent relationships.

Therefore, the relation has the join dependency:

(MovieID, ActorID, DirectorID, PlatformID)

which means that the original relation can be reconstructed by joining smaller relations representing these independent associations.

3. 5NF Requirement

A relation is in **5NF** if every non-trivial join dependency is implied by the candidate keys of the relation.

The original relation contains independent relationships among:

- Movies and Actors
- Movies and Directors
- Movies and Streaming Platforms

Therefore, these relationships should be separated.

4. Decomposition

Movie

Movie(MovieID, MovieTitle)

- Primary Key: MovieID

Actor**Actor(ActorID, ActorName)**

- Primary Key: ActorID

Director**Director(DirectorID, DirectorName)**

- Primary Key: DirectorID

Platform**Platform(PlatformID, PlatformName)**

- Primary Key: PlatformID

MovieActor**MovieActor(MovieID, ActorID)**

- Composite Primary Key: (MovieID, ActorID)

MovieDirector**MovieDirector(MovieID, DirectorID)**

- Composite Primary Key: (MovieID, DirectorID)

MoviePlatform**MoviePlatform(MovieID, PlatformID)**

- Composite Primary Key: (MovieID, PlatformID)

5. Final 5NF Schema

Relation	Attributes	Primary Key
Movie	MovieID, MovieTitle	MovieID
Actor	ActorID, ActorName	ActorID
Director	DirectorID, DirectorName	DirectorID
Platform	PlatformID, PlatformName	PlatformID
MovieActor	MovieID, ActorID	(MovieID, ActorID)
MovieDirector	MovieID, DirectorID	(MovieID, DirectorID)

Relation	Attributes	Primary Key
MoviePlatform	MovieID, PlatformID	(MovieID, PlatformID)

6. Lossless-Join Verification

The decomposition is lossless because the original information can be reconstructed using natural joins on the common key attributes.

For example:

Movie $\bowtie$ MovieActor $\bowtie$ MovieDirector $\bowtie$ MoviePlatform

can reconstruct the movie-related relationship.

The descriptive information can then be obtained by joining with:

Actor, Director, and Platform

Thus:

Movie $\bowtie$ MovieActor $\bowtie$ MovieDirector $\bowtie$ MoviePlatform $\bowtie$ Actor $\bowtie$ Director $\bowtie$ Platform

reconstructs the original information without introducing spurious tuples, provided the assumed independence of the three relationships is valid.

7. Why 5NF is Achieved

Each final relation represents a single independent relationship:

- **MovieActor** represents movie–actor participation.
- **MovieDirector** represents movie–director association.
- **MoviePlatform** represents movie availability on a platform.

No further non-trivial join dependency exists that can be decomposed without losing information.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand